ETH zürich

MAN Energy Solutions

Town Meeting

How to Achieve the Greatest Economic Benefit with Mega Heat Pumps

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Welcome to this Board Meeting

Consultant: Tobi

Consultant: Sevi

Why are we here today?

1. Townhall meeting with stakeholders
2. Change heating system to a more r

Townhall Meeting

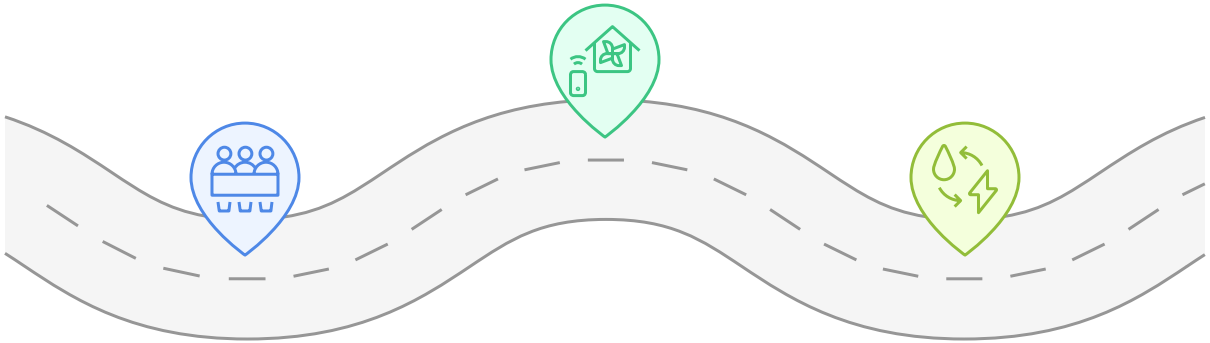
Gathering stakeholders for discussion

Discuss Renewable Heating

Exploring options for a renewable heating system

Decide on Change

Making a decision to switch to a renewable system



Esbjerg

Huge heat pump powered by wind farm 75% renewables in danmark

Population: 72‘000

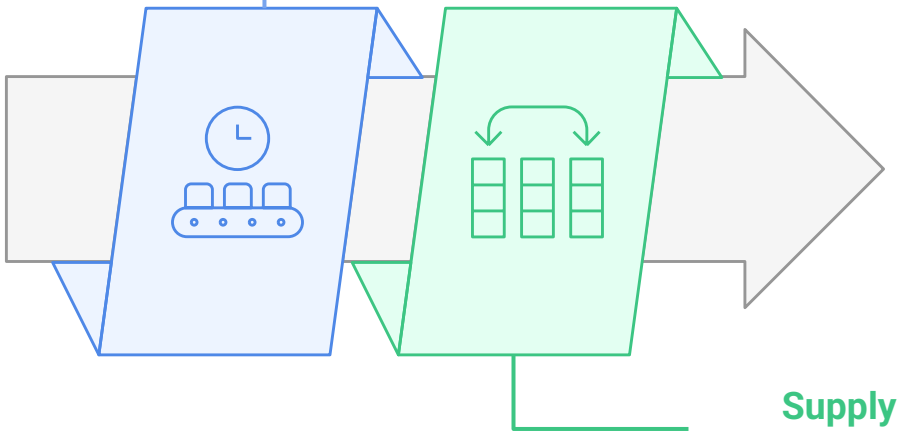
Demand

How does FCR work?

Balanced 4h Blocks

The system divides time into balanced four-hour blocks for energy management.

1. explain balanced 4h blocks
2. storage -> supply follows demand

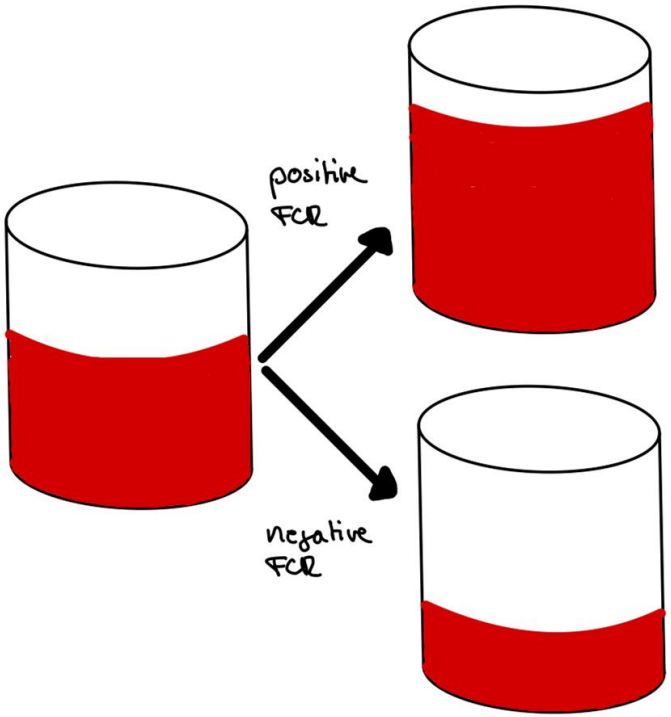


Supply Follows Demand

Energy storage is adjusted to meet the fluctuating energy demand.

Thermal Storage

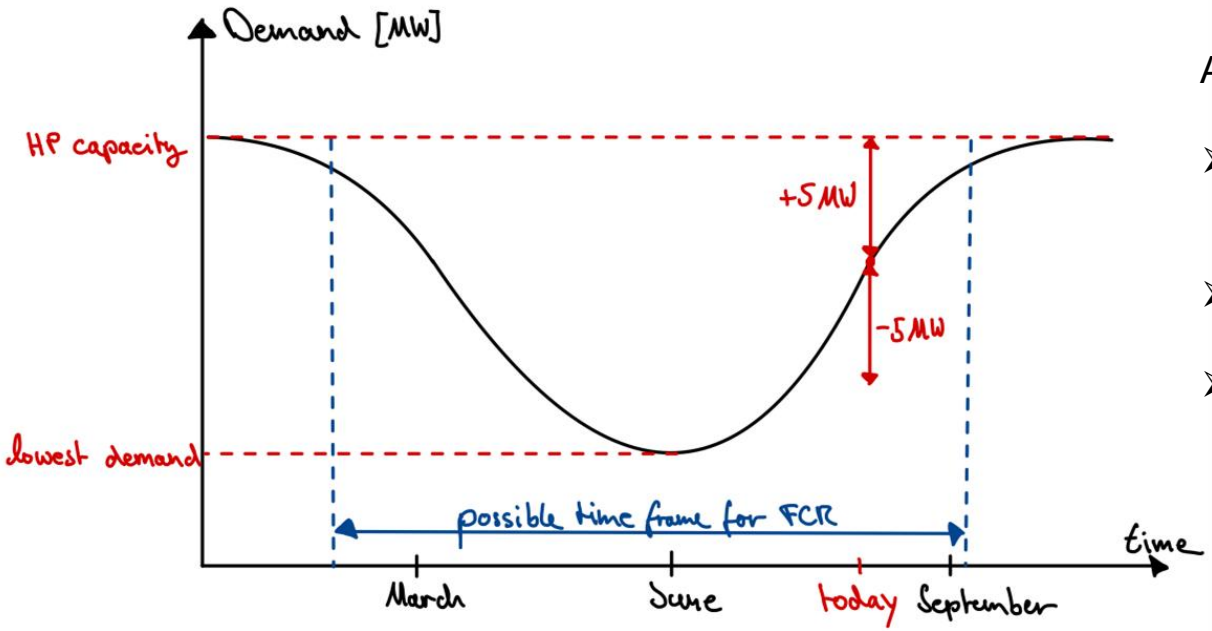
$$2 * 2h * (HP\ Capacity - lowest\ Demand) = 2 * 2h * FCR\ Capacity = Storage\ Size\ in\ MWh$$



Assumptions:

- Positive and negative FCR activations balance out over each 4-hour block
- Thermal storage is initially “filled” to 50%
- Worst case is 2h full positive or negative activation

How to Provide FCR Service with a Mega Heat Pump?

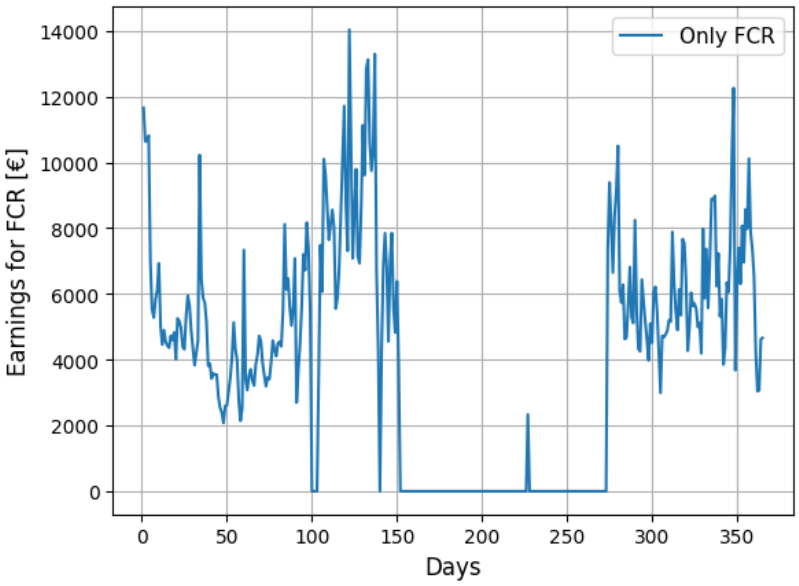


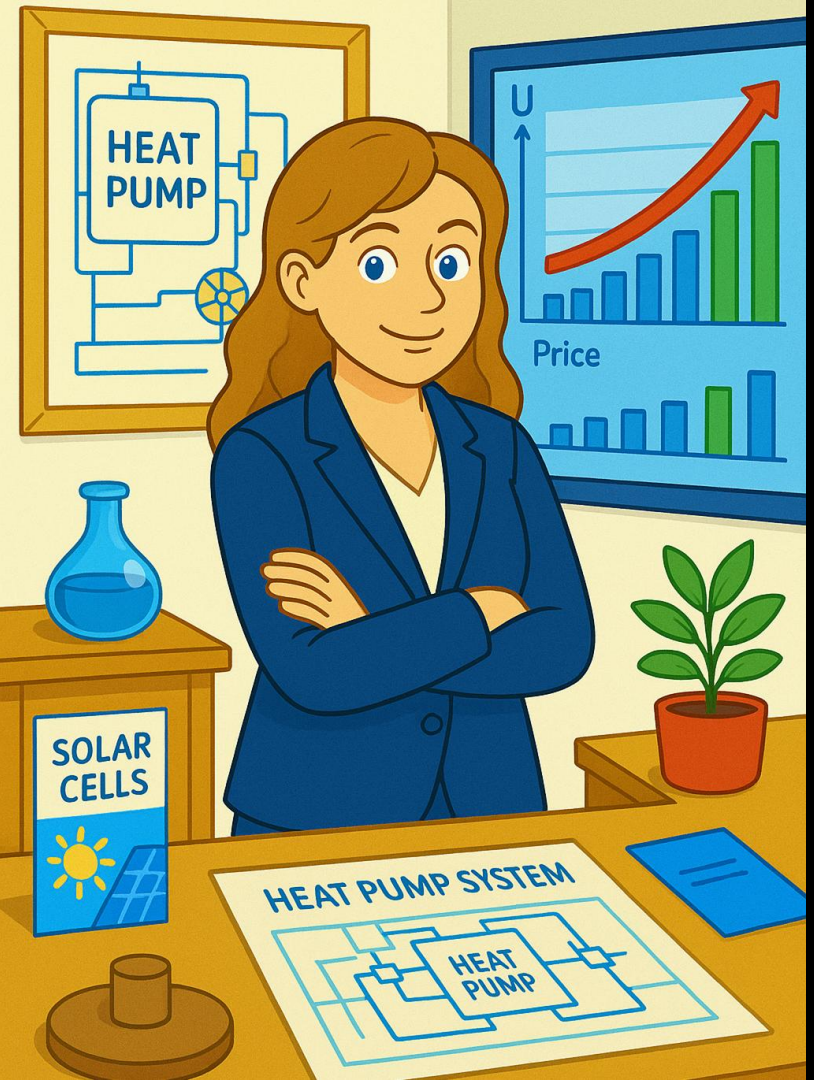
Assumptions: -> strong statements

- if demand < nominal capacity, excess capacity can provide FCR
- Thermal storage is big enough!
- Positive and negative FCR activations balance out over each 4-hour block!

Economic Benefit of Providing FCR

1. Plot with only base case and providing FCR without oversizing



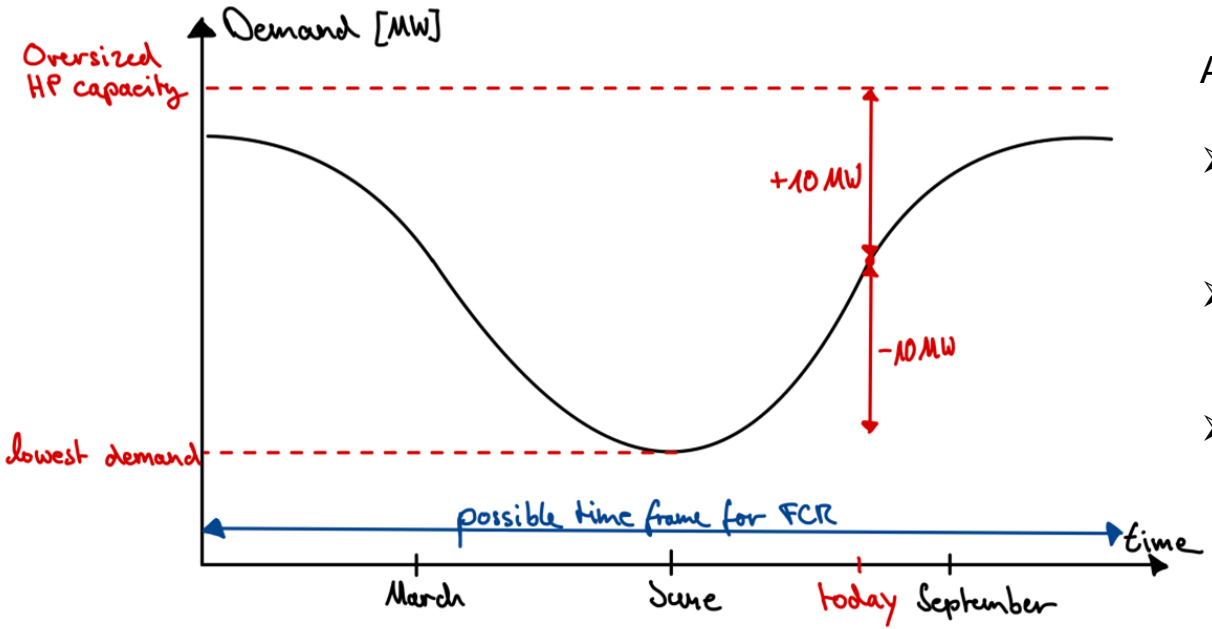


Investor: Kim

How can I maximize the profit I get out of my heat pump?

Mega Heat Pump with Oversizing

How can I maximize the profit I get out of my heat pump?

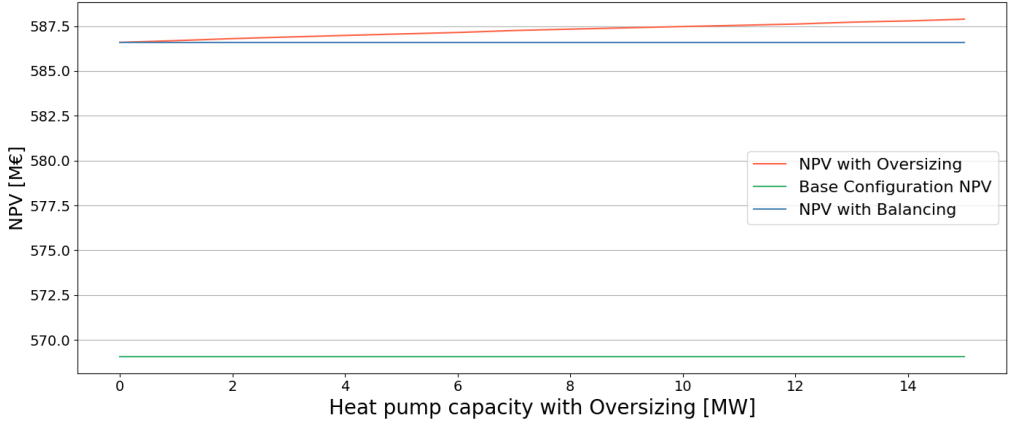


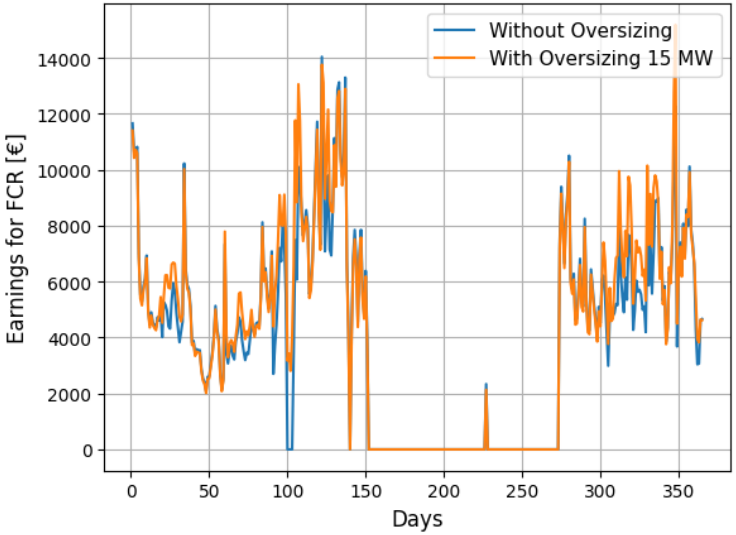
Assumptions:

- if demand < nominal capacity, excess capacity can provide FCR
- Sufficiently large thermal storage to handle fluctuations
- Positive and negative FCR activations balance out over each 4-hour block

Economic Benefit of FCR with Oversizing

1. Plot with all cases
2. Plot with daily FCR showing that gap in summer





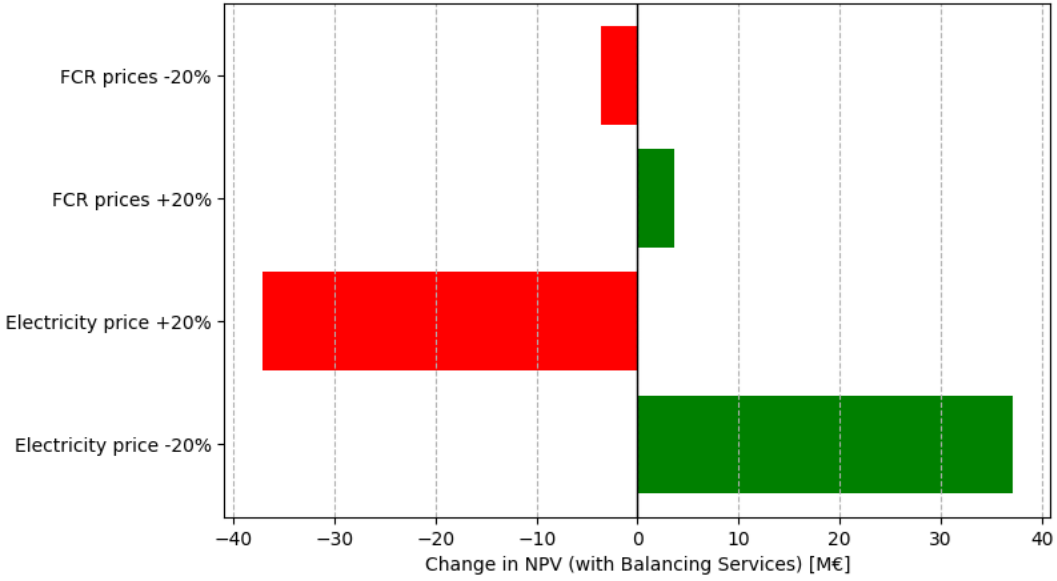
Mayor: Beltran

This revenue seems nice, but how will it change if prices change?

Sensitivity Analysis

This revenue seems nice, but how will it change if prices change?

1. Tornado Plot
2. Table with numbers



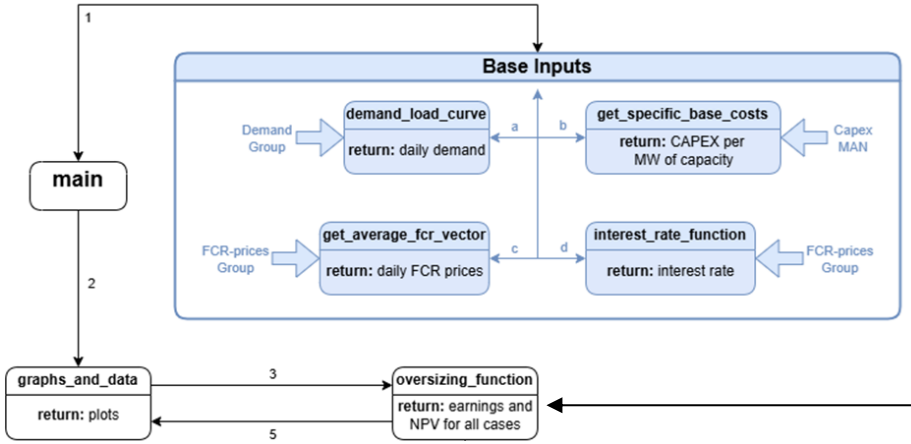
Scepticist: Till

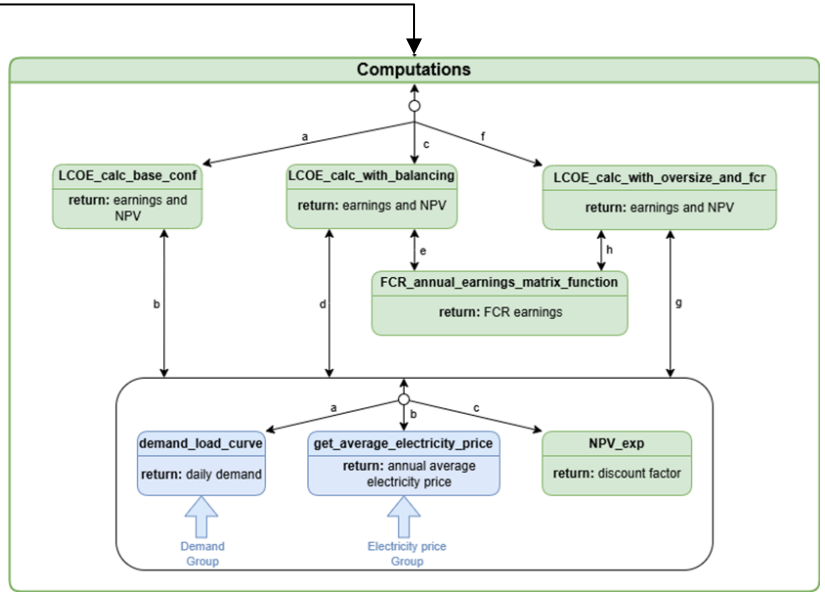
How did you get to those values? I don’t believe them!

Model Structure

How did you get to those values? I don't believe them!

- Flowchart
- Inputs



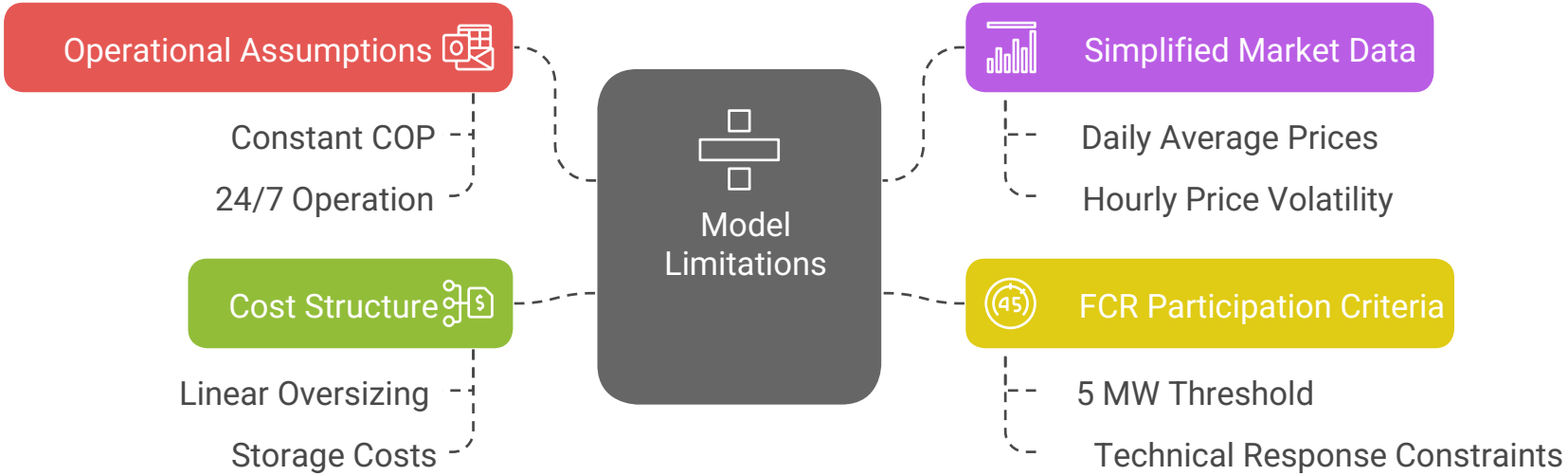


Citizen: Con

This all sounds great, but I’m a bit confused. So, what is the best option now?

Summary

I'm a bit confused.
So, what is the
best option now?

Thank you for this productive Meeting!
MAN

Limitations of Model